Practice workbook · Arithmetic: standard form, estimates and rounding

Totals, differences, estimates, standard form, decimal places and significant figures.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-19

1. Question 5 · part f · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part f only.

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5 Rocio conducted a case study on her patient who had an addiction to alcohol. She worked with the patient for six months to help treat their addiction.

(a) Define what is meant by a 'case study'.

(1)

(b) Rocio recorded the conversations with the patient as part of her case study. She also monitored the alcohol consumption of the patient in units of alcohol (one unit of alcohol = 10 ml).

Her results are shown in **Table 2**.

Month	Total alcohol consumption in units
March	298
April	221
May	181
June	140
July	99
August	59

Table 2

Describe, using an example from Rocio's case study, what is meant by 'quantitative data'.

(2)

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1. Question 5 · part f · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part f only.

(c) Estimate the number of units of alcohol that the patient has reduced their consumption by between March and August.

(1)

Estimation .....

(d) For the months of March and April, Rocio recorded the patient's alcohol consumption each week. Her results are shown in **Table 3**.

Week	Total alcohol consumption in units
March Week 1	79
March Week 2	77
March Week 3	72
March Week 4	70
April Week 1	79
April Week 2	63
April Week 3	45
April Week 4	34

Table 3

Calculate the mode for the units of alcohol consumed by the patient using **Table 3**.

(1)

Mode .....

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1. Question 5 · part f · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part f only.

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(e) Calculate the median score for the units of alcohol consumed by the patient using **Table 3**.

(2)

Median .....

(f) On one day in March, the patient drank a total of 12.81924 units of alcohol.

Give the units of alcohol to **two** decimal places.

(1)

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(Total for Question 5 = 8 marks)

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Working and answer

2. Question 3 · part a · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part a only.

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3 Vanessa investigated age differences in driving speed. She conducted an observation by a road in the local town. Vanessa recorded the speed that young drivers and mature drivers drove their cars in miles per hour (mph).

The number of young drivers and mature drivers who were recorded driving at each speed is shown in **Table 3**.

	1mph to 10mph	11mph to 20mph	21mph to 30mph	31mph to 40mph
Young driver	1	4	9	12
Mature driver	1	5	14	18

Table 3

(a) Calculate the difference between the number of young drivers who were recorded driving at 1mph to 20mph and the number of young drivers who were recorded driving at 21mph to 30mph.

(1)

Difference .....

(b) Calculate the ratio of young drivers to mature drivers that Vanessa recorded driving at 31mph to 40mph.

(1)

Ratio .....

(c) Explain **one** improvement that Vanessa could make to her investigation.

(2)

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(Total for Question 3 = 4 marks)

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2. Question 3 · part a · 1 mark · Working and answer

Working and answer

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3. Question 6 · part a · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part a only.

6 Kayla investigated whether the use of positive facial expressions (such as smiles) by actors in television advertisements resulted in people being more likely to buy the product being advertised.

First, she analysed two newspaper articles about the use of facial expressions in television advertisements.

Kayla then used an online questionnaire to find out if facial expressions influenced the purchase of a particular brand of cereal. 39,980,000 responses were collected in total.

(a) Give 39,980,000 in standard form.

(1)

39,980,000 in standard form .....

(b) 10,003,010 responses showed positive facial expressions used by actors in television advertisements did not influence their decision to buy the cereal.

Estimate the number of responses that showed positive facial expressions **did** influence their decision to buy the cereal.

(1)

Estimation .....

(c) 63% of the total responses to the online questionnaire were completed by families with children.

Calculate the number of families with children who completed the online questionnaire.

(1)

Number of families with children who completed the online questionnaire .....

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3. Question 6 · part a · 1 mark · Working and answer

Working and answer

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4. Question 6 · part b · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b only.

6 Kayla investigated whether the use of positive facial expressions (such as smiles) by actors in television advertisements resulted in people being more likely to buy the product being advertised.

First, she analysed two newspaper articles about the use of facial expressions in television advertisements.

Kayla then used an online questionnaire to find out if facial expressions influenced the purchase of a particular brand of cereal. 39,980,000 responses were collected in total.

(a) Give 39,980,000 in standard form.

(1)

39,980,000 in standard form .....

(b) 10,003,010 responses showed positive facial expressions used by actors in television advertisements did not influence their decision to buy the cereal.

Estimate the number of responses that showed positive facial expressions **did** influence their decision to buy the cereal.

(1)

Estimation .....

(c) 63% of the total responses to the online questionnaire were completed by families with children.

Calculate the number of families with children who completed the online questionnaire.

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Number of families with children who completed the online questionnaire .....

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4. Question 6 · part b · 1 mark · Working and answer

Working and answer

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5. Question 3 · part e · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

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3 Becca wants to find out if there are sex differences in male and female brain functioning. She decides to use a laboratory experiment.

(a) State **one** feature of a 'laboratory experiment'.

(1)

(b) Describe how Becca could use volunteer sampling to gather the participants for her laboratory experiment.

(2)

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(c) Becca will give each participant a coordination task and a word-meaning task to test for sex differences in brain functioning. She decides to use a single-blind technique.

Describe **one** reason why Becca used a single-blind technique for her experiment.

(2)

(d) Becca conducted her laboratory experiment with 30 male and 25 female participants.

Calculate the ratio of male to female participants in Becca's sample.

(1)

Ratio .....

5. Question 3 · part e · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

(e) Becca's results are shown in **Table 2**.

	Mean time taken to accurately complete the coordination task (seconds)	Mean time taken to accurately complete the word-meaning task (seconds)
Males	57.24	63.51
Females	64.43	51.36

Table 2

Calculate the difference between the mean time (seconds) taken by **males** to accurately complete the coordination task and word-meaning task.

You must give your answer to two significant figures.

(2)

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Difference in mean time (seconds) taken by males .....

(f) Explain **one** conclusion that Becca could make from her results.

(2)

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(Total for Question 3 = 10 marks)

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5. Question 3 · part e · 2 marks · Working and answer

Working and answer

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6. Question 3 · part b · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part b only.

3 Ryan used a questionnaire to assess whether students had a fixed or growth mindset in four secondary schools. He asked the students to complete the questionnaire online.

His results are shown in **Table 2**.

School	Number of students with a fixed mindset	Number of students with a growth mindset
A	458	931
B	211	439
C	641	838
D	189	271

Table 2

(a) Calculate the mean number of students with a fixed mindset.

(1)

Mean .....

(b) Calculate the total number of students who completed the questionnaire.

(1)

Total .....

(c) Calculate the range for the number of students with a growth mindset.

(1)

Range .....

(Total for Question 3 = 3 marks)

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6. Question 3 · part b · 1 mark · Working and answer

Working and answer

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7. Question 5 · part c · 1 mark

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Further practice · Attempt part c only.

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7. Question 5 · part c · 1 mark

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Further practice · Attempt part c only.

(c) Estimate the number of units of alcohol that the patient has reduced their consumption by between March and August.

(1)

Estimation .....

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7. Question 5 · part c · 1 mark · Working and answer

Working and answer

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